WCAG Accessibility Report

Generated: 2026-07-31 12:45 | Branch: Dev | Commit: 2ba35c431 | Framework: Angular 20 + Angular Material 20 | Target: src/app/modules/laboratory-functions/component-production/cmci | Iterations: 1

Accessibility Score

WCAG 2.1 Level AA | Excellent

95

Summary

0	0	0	1
Critical	Serious	Moderate	Minor

Iteration	Issues Found	Issues Fixed	Remaining	Review Outcome
1	11	10	1 (intentional)	Passed QA (95/100)

Resolved (10 issues fixed in 1 iteration)

Serious **S1: Duplicate aria-label on mat-select — WCAG 4.1.2**

File: convert-modify-components.html:18, 46
Fix: Removed redundant aria-label from Product Type and Attribute Code mat-select elements. The mat-label already provides the accessible name.

Serious **S2: Display cards lack semantic structure — WCAG 1.3.1**

File: convert-modify-components.html:38-72, 146-155
Fix: Added role="group" with aria-label and aria-labelledby to all 5 read-only display cards (Description, Attribute Group, Derived Product Code, Short Name).

Moderate **M1: Filter form has no accessible name — WCAG 1.3.1**

File: convert-modify-components.html:13
Fix: Added aria-label="Filter by Product Type and Attribute Code".

Moderate **M2: Entry form has no accessible name — WCAG 1.3.1**

File: convert-modify-components.html:91
Fix: Added aria-label="Component entry details".

Moderate **M3: Duplicate IDs on process date errors — WCAG 4.1.1**

File: convert-modify-components.html:180-188
Fix: Gave each mat-error a unique ID (process-date-required, process-date-future, process-date-min). Updated aria-describedby to conditionally reference the active error.

Moderate **M4: Warning-error color fails contrast — WCAG 1.4.3**

File: convert-modify-components.scss:233
Fix: Changed color from #ff9800 (2.88:1 ratio) to #9a6700 (~5.2:1 ratio), meeting the 4.5:1 minimum.

Moderate **M5: Unused ARIA live region — WCAG 4.1.3**

File: convert-modify-components.html:1-2
Fix: Removed the unused aria-live announcer div. MatSnackBar provides built-in ARIA live announcements via aria-live="assertive".

Moderate **M6: No focus-visible on custom buttons — WCAG 2.4.7**

File: convert-modify-components.scss:180-246
Fix: Added :focus-visible { outline: 2px solid #228189; outline-offset: 2px; } to reprint, clear, and clear-filters buttons.

Minor **m1: Product options lack combined aria-label — WCAG 1.3.1**

File: convert-modify-components.html:129
Fix: Added [attr.aria-label]="product.code + ' - ' + product.description" to mat-option elements.

Minor **m3: Empty state message misleading — WCAG 3.3.2**

File: convert-modify-components.html:308-309
Fix: Changed "No Results to Display / Please enter search criteria" to "No Entries Yet / Scan a unit ID and fill in the entry fields to process a component change".

Can Do Better

Your accessibility score is **95/100**. The critical and serious issues have been resolved. The following enhancements will push your score closer to 100 and improve the experience for users of assistive technologies.

> **autocomplete="off" on forms — WCAG 1.3.5**

Both forms disable browser autocomplete. This is intentional for the barcode scanning workflow to prevent browser interference with scanned values. If non-scan fields are added in the future, consider enabling autocomplete selectively.

Screen reader testing for dynamic content — WCAG 4.1.3

Verify that MatSnackBar notifications (entry saved, validation errors) are properly announced by screen readers. Consider adding a custom ARIA live region if snackbar announcements are missed in production testing.

Keyboard shortcut documentation — WCAG 2.1.4

The barcode scanning fields accept Enter key to submit. Consider providing keyboard shortcut documentation or tooltips for screen reader users who may not discover this interaction pattern.

Data table accessibility — WCAG 1.3.1

The app-data-table shared component handles table accessibility. Verify that column headers use <th scope="col">, sortable columns announce sort state, and pagination controls are labeled. These are shared component concerns.

Error recovery guidance — WCAG 3.3.3

When verify product doesn't match the derived product code, the field is cleared and refocused. Consider adding an inline mat-error message explaining the mismatch, in addition to the snackbar notification, so the error persists visually.

Recommended Manual Validation

- ☐ Navigate the entire page using only the keyboard (Tab, Shift+Tab, Enter, Arrow keys)
- ☐ Test with a screen reader: NVDA (Windows), VoiceOver (macOS/iOS), or TalkBack (Android)
- ☐ Verify all form error messages are announced by screen readers
- ☐ Check color contrast for custom colors using the WebAIM Contrast Checker
- ☐ Validate dynamic content updates are announced via ARIA live regions
- ☐ Run axe-core browser extension for any pages not covered by the automated scan
- ☐ Test the barcode scanning workflow with assistive technology enabled
- ☐ Verify the data table component announces sort changes and page navigation

Angular Material Accessibility Notes

mat-select labeling

Angular Material's mat-select inside mat-form-field is automatically labeled by mat-label. Do not add redundant aria-label attributes — they override the visible label association.

Unique ID generation

Use a static counter pattern (componentId + name) to generate unique IDs for form elements. This prevents ID collisions when multiple instances of the same component exist on a page.

MatSnackBar announcements

MatSnackBar uses aria-live="assertive" by default. Verify this works with your target screen readers before adding custom live regions. Custom live regions may conflict with MatSnackBar's built-in accessibility.

Conditional mat-error IDs

When multiple mat-error elements exist for the same field (e.g., required, min, max), give each a unique ID and wire aria-describedby conditionally to reference only the currently active error.
